Local orders at a simple finite branch of weighted Proj

Formal proof packet

27 October 2026

This packet records a conditional geometry proof. It adds no target experiment or cost ratio. The editable graph, gate CSV, Lean source, and compiler log are beside this PDF.

Checked statement

For a field K, quintic f, and rational a with $f(a) = 0$ and $f'(a) \neq 0$, the structure-sheaf stalk of the actual weighted-Proj curve at the finite branch point is a discrete valuation ring. Under the checked stalk isomorphism with the affine branch localization, the pullback of y has order one and the pullback of $x - a$ has order two.

Rational simple branch on actual weighted Proj

$\Downarrow$ checked structure-sheaf stalk isomorphism

Affine branch localization, a DVR with $y^2 = (x - a)q$ and q a unit

$\Downarrow$ checked local valuation transport

$\operatorname{ord}(y) = 1, \qquad \operatorname{ord}(x - a) = 2$

The DVR criterion is discussed in the Stacks Project, Lemma 10.119.7. The checked transport uses the finite-branch stalk isomorphism and the repository's affine branch-domain and local-valuation lemmas. The four Lean theorem names and source bindings are in `finite-branch-dvr.csv`.

Verification and remaining gates

Direct Lean 4.34.1 compilation with Mathlib `d13f23b` exited zero. All four theorem axiom audits list only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Lean source SHA-256:

`181f480366672eadf9fa0e08fac1786f467789d71f677ea858aadac6252cb35f.`

The proof does not establish every numerator factorization or nonbranch zero multiplicity, a global principal divisor, a Pic^0 class equation, or semantics for every compiled Rust branch. The frozen 1,260-target rows and matched-rho cost ratios are unchanged.